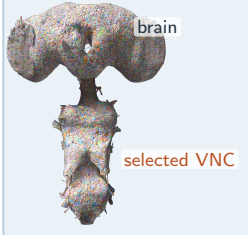
Measured wiring to a non-fly body

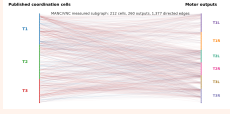
Anatomy



MaleCNS / VNC context

1. Freeze VNC

11,000 model nodes
485,074 signed edges



filter ≥ 5 | signed edges

named descending-input drive
weights stay fixed

2. Map signals

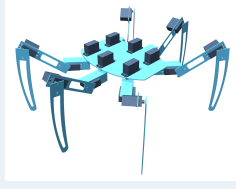
260 motor channels

combine opposing signals

18 joint targets

fixed map
no learned conversion

3. Simulate



MuJoCo | 6 legs | 18 joints
home pose; PD joint control

4. Evaluate

Localized -0.0112 m
T2 segment lesion

Graded $\rho = -0.886$
6 rewiring fractions

Across nine body tests
0 of 12 controls beat it

drive contrasts; gait from CPG